

Deep Learning for Natural Language Processing

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GIAN Course on “*AI for Data Representation, Analytics, and Visualization*”

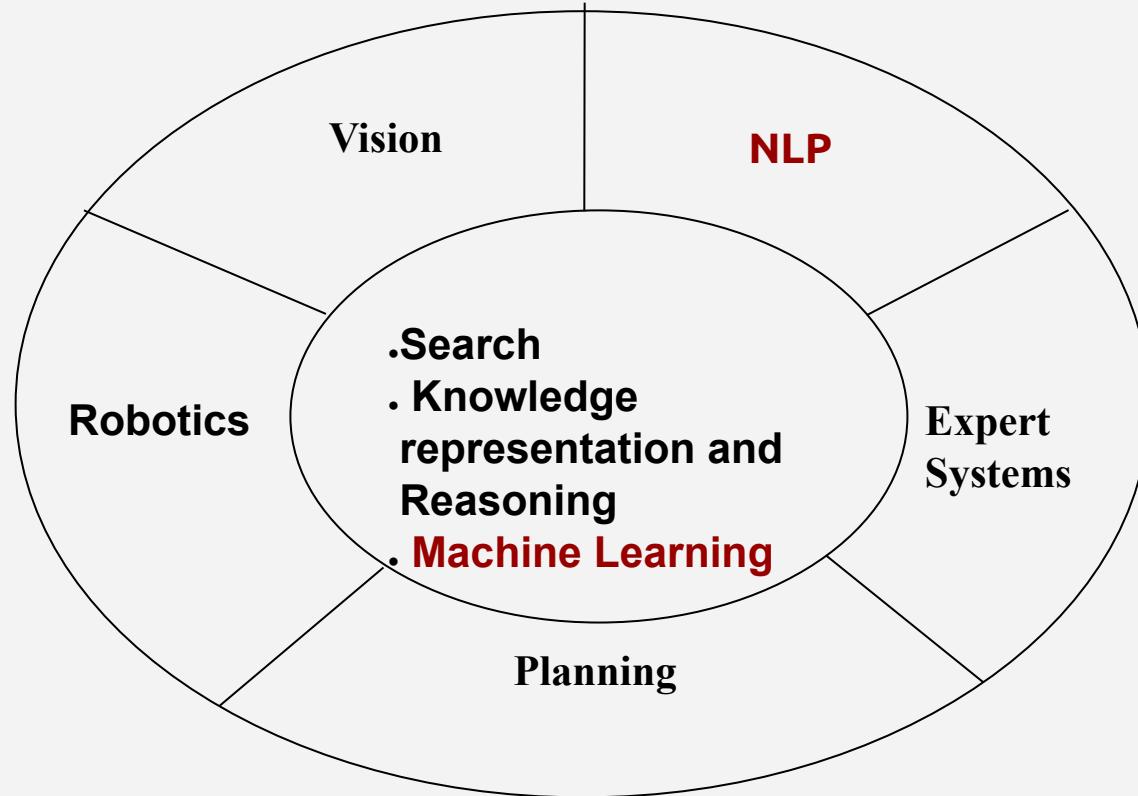
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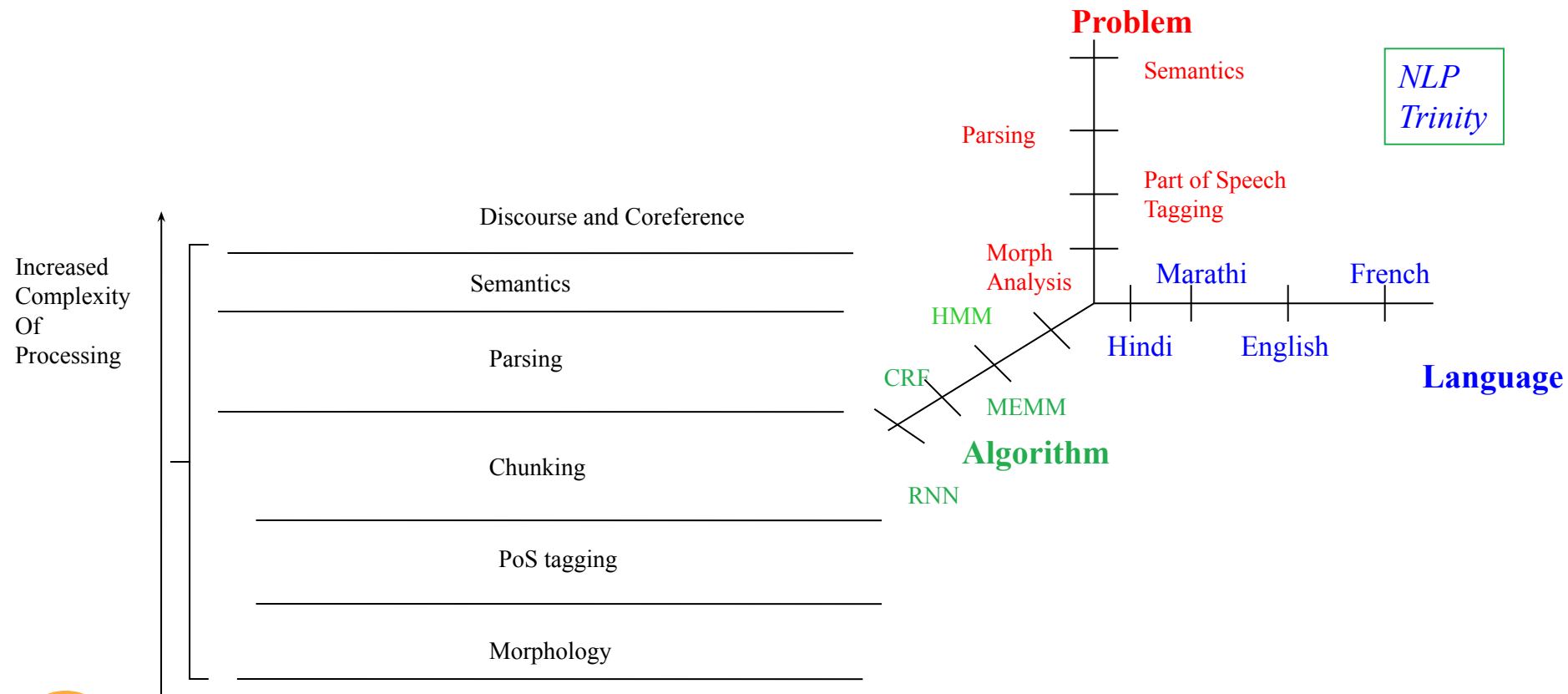
Outline

- **AI-NLP-ML**: Background
- Some interesting applications
 - Fine-grained Sentiment Analysis: **Aspect based Sentiment Analysis (ABSA)**
 - **Multimodal Sentiment and Emotion Analysis**
 - **Conversational AI- Empathetic Agent**
 - **Machine Translation**
 - **AI in Scholarly Communication**
- **AI for Social Good-** Conversational Agent for assisting crime victims
- **Takeaways**

AI: The various Components



NLP Trinity



Multilinguality: Indian situation

- **Major streams**

Indo European

Dravidian

Sino Tibetan

Austro-Asiatic

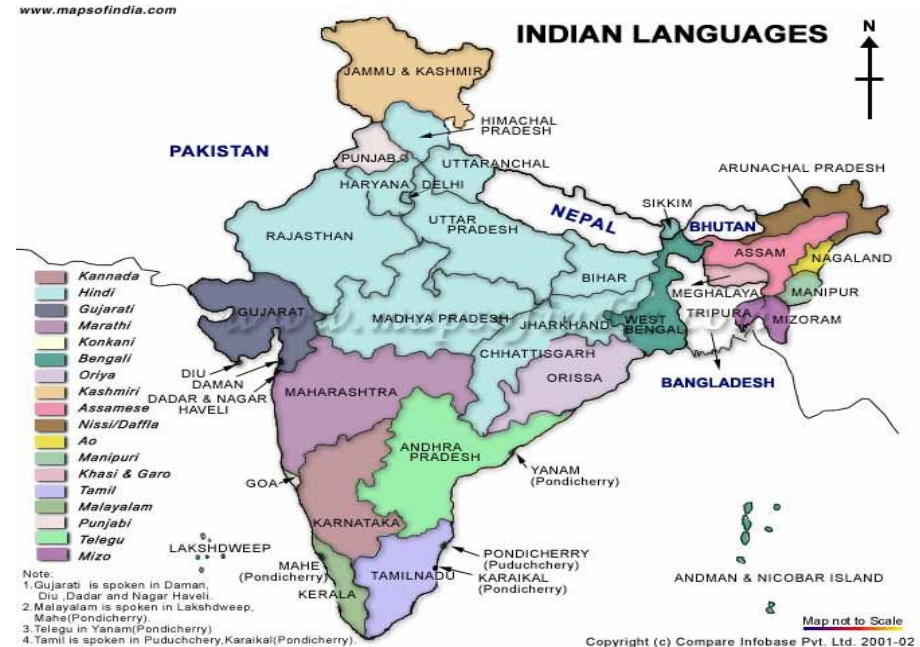
- Some languages are ranked within 20 in the world in terms of the populations speaking them

Hindi :3rd (~698 milion)

Bangla: 7th (~300 million)

Urdu : 11th (~170 million)

Marathi 20th (~74 million)



Background: Indian Context

- India is a multi-lingual country with great linguistic and cultural diversities
- 22 official languages mentioned in the Indian constitution
- However, Census of India in 2001 reported-
 - **122 major languages**
 - **1,599 other regional languages**
 - **13 scripts**
 - **720 dialects**
 - **30 languages** are spoken by more than **one million native speakers**
 - **122** are spoken by more than **10,000 people**
- **20%** understand English
- **80%** cannot understand

Background

- Phenomenal growth in the number of internet users, social media (**Facebook, Twitter** etc.)
- Increasing tendency of using Indian language contents for exchanging information
- **Digital divide** cannot be tackled unless citizens are given flexibility in **communicating in their own languages**

Language Technology or Natural Language Processing (NLP) that deals with developing theories and techniques for effective communication in human languages play an important role towards creating this digital society

AI, NLP, ML: From Past to Present to Future (1/3)

- **AI/NLP based systems have enabled wide-range of applications**
 - Google's powerful search engines
 - Alexa, Google's MT
 - Several systems related to healthcare, mental health, processing social media texts etc.
 - Fake news detector and so on
 - Chess playing
 - & NOW **THE ChatGPT!!, GPT-4 and DeepSeek!!**
- **Shallow ML algorithms (corresponds to Statistical NLP)**
 - Used extensively (HMM, MaxEnt, CRF, SVM, Logistic Regression etc.)
 - Requires handcrafting of features
 - Time-consuming
 - Curse of dimensionality (because of joint modeling of language models)

AI, NLP and ML: From Past to Present to Future (2/3)

- **Deep Learning algorithms**

- No feature engineering
- Success of distributed representations (*Neural language models*)

- **Some recent developments**

- The rise of distributed representations (e.g., Word2vec, GLOVE, ELMO, BERT etc)
- Convolutional, recurrent, recursive neural networks, Transformer, Reinforcement learning
- Unsupervised sentence representation learning
- Combining deep learning models with memory-augmenting strategies

- **Explainable AI**

Research Areas

Conversational AI

- NLU
- Empathetic dialogue generation
- Knowledge grounded dialogue generation
- Personalization

Machine Translation

- Judiciary
- Education
- E-commerce

- Artificial Intelligence
- **Natural Language Processing**
- Machine Learning Applications
- Text Mining

Sentiment & Emotion Analysis

- Coarse-grained & Fine-grained
- Multilingual Sentiment Analysis
- Multimodal Sentiment & Emotion Analysis

Information Extraction

- QA
- Event extraction
- Misinformation detection
- Hate speech detection
- Novelty detection

Socially relevant problems

Industry relevant problems

Flipkart, Wipro, Samsung, Accenture, LG Soft, Elsevier etc

Today's Conversational AI: **A Long way from ELIZA**

- **Generating coherent and engaging responses in conversations**
 - Through Deep Language Understanding and Reasoning
- Should understand a user's need, context and mood
- Should be able to respond with **personalization, sentimental and emotional analysis**
- Balancing human-like aspects such as **specificity** and **empathy**

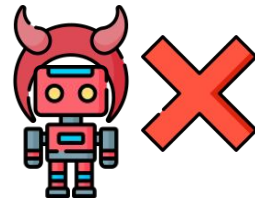
*The chatbot market size is projected to grow from \$2.6 billion in 2019 to \$9.4 billion by 2024 at a compound annual growth rate (**CAGR**) of 29.7% (**BusinessInsider**)*

Empathetic Conversational AI



I finally got promoted today at work!!

Why would anyone promote you?



Congrats!! That's great!!

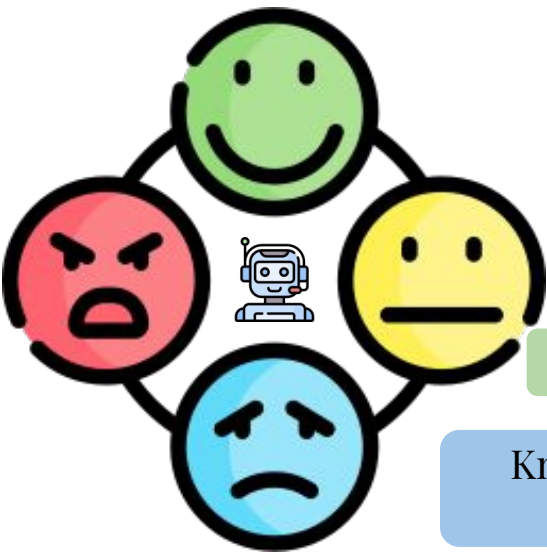
Empathy

Multi-Emotion Controlled Dialogue Generation (AAAI 2021)

It's amazing, I am thrilled you got promoted [Surprise, Joy] [0.3, 0.9]

Stop sulking, I am sure you will manage it [Anger, Acceptance] [0.7, 0.4]

I am sorry this could be an infection or cancer [Sadness, Fear] [0.6, 0.3]



Sentiment awareness

Knowledge grounded
Conversation

Intent

Context

Mood

Aware
&
Relatable

Politeness and Personalization (HLT/NAACL 2019, INLG 2021)

We are sorry to see you are having trouble. Help has arrived. How can I help?

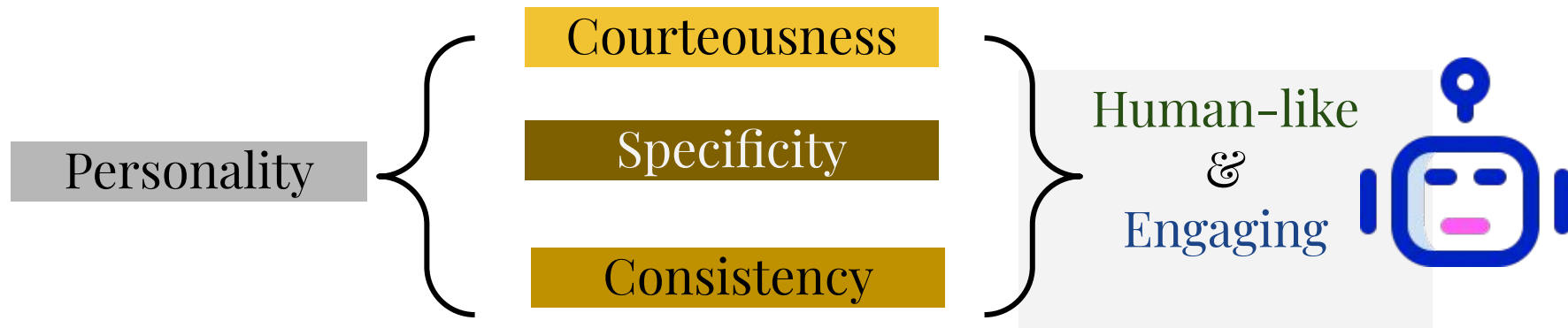
Apology

Hey Christiano!! Happy to help. Can you send us a screenshot?

Greet, Assurance

Thank you for your kind words!! Enjoy your show while flying!!

Appreciation

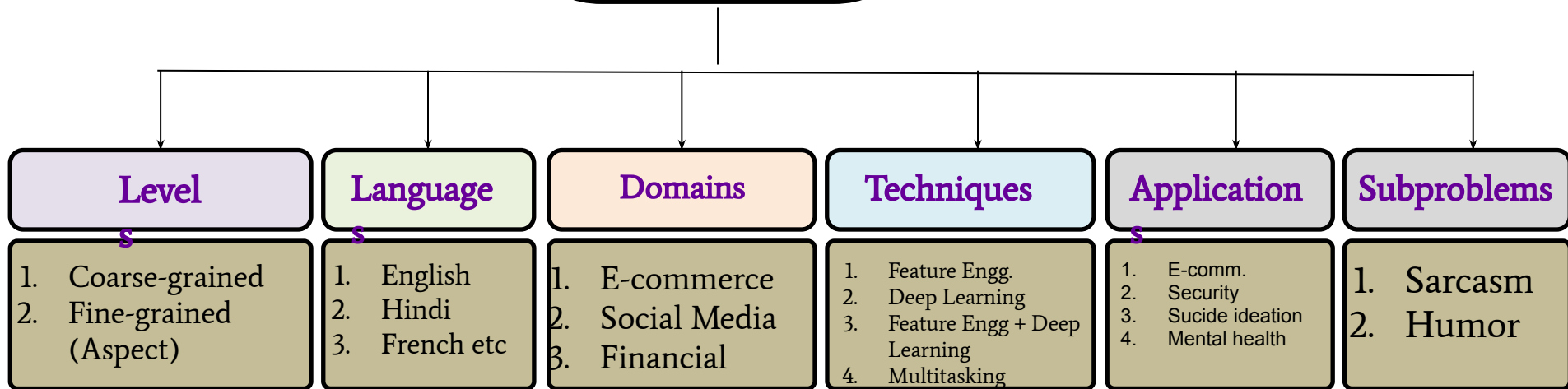


Sentiment Analysis

Publications:

IEEE Transaction on Affective Computing, ACM TKDD, Neurocomputing, IEEE Computational Intelligence, ACM TALLIP, ACL, COLING, HLT/NAACL, EMNLP etc

Sentiment Analysis



Aspect based Sentiment Analysis: *More informed Decision*

The **zoom** is excellent, but the **LCD** is blurry.
Great value for the **price**.
Although the **display** is poor the **picture quality** is amazing.
Batteries drain pretty quickly.
I love this camera but for short **battery life** is definitely a pain.
It is a good camera for the **price**.

Sentence-level SA

• **Product X:** **3.5 ★**

Aspect-level SA

- Price: 4.5 ★
- Zoom: 5.0 ★
- Picture Quality: 4.5 ★
- Battery: 1.5 ★
- Display: 1.0 ★

- *Low-resource settings*
- *Multilingualism*
- *State-of-the-art performance*
- *Resource creation*

Multi-tasking and Multimodality

- Sentiments ([Pang et al., 2005](#)) and emotions ([Ekman, 1999](#)) are closely related
 - *anger, fear, disgust, sad* → **Negative**
 - *happy, surprise* → **Positive**
- Each task improves its performance in association with the other participating tasks
- Multitasking offers **reduced complexity** because a single system can handle multiple problems/tasks at the same time

Utterance	Affect	Text	Acoustic	Visual
<i>Thanks for putting me on hold! I have all the time in the world.</i>	Sentiment (Negative)	-	Intensity, Pitch, Tone etc.	Gesture, eyes movement etc.
<i>I'm fine.</i>	Emotion (Sad)	-		Facial expression

Pang et al., 2005- Seeing stars: Exploiting class relationships for sentiment categorization with respect to rating scales.
Ekman, 1999- The handbook of cognition and emotion.

Artificial Intelligence in Peer Review

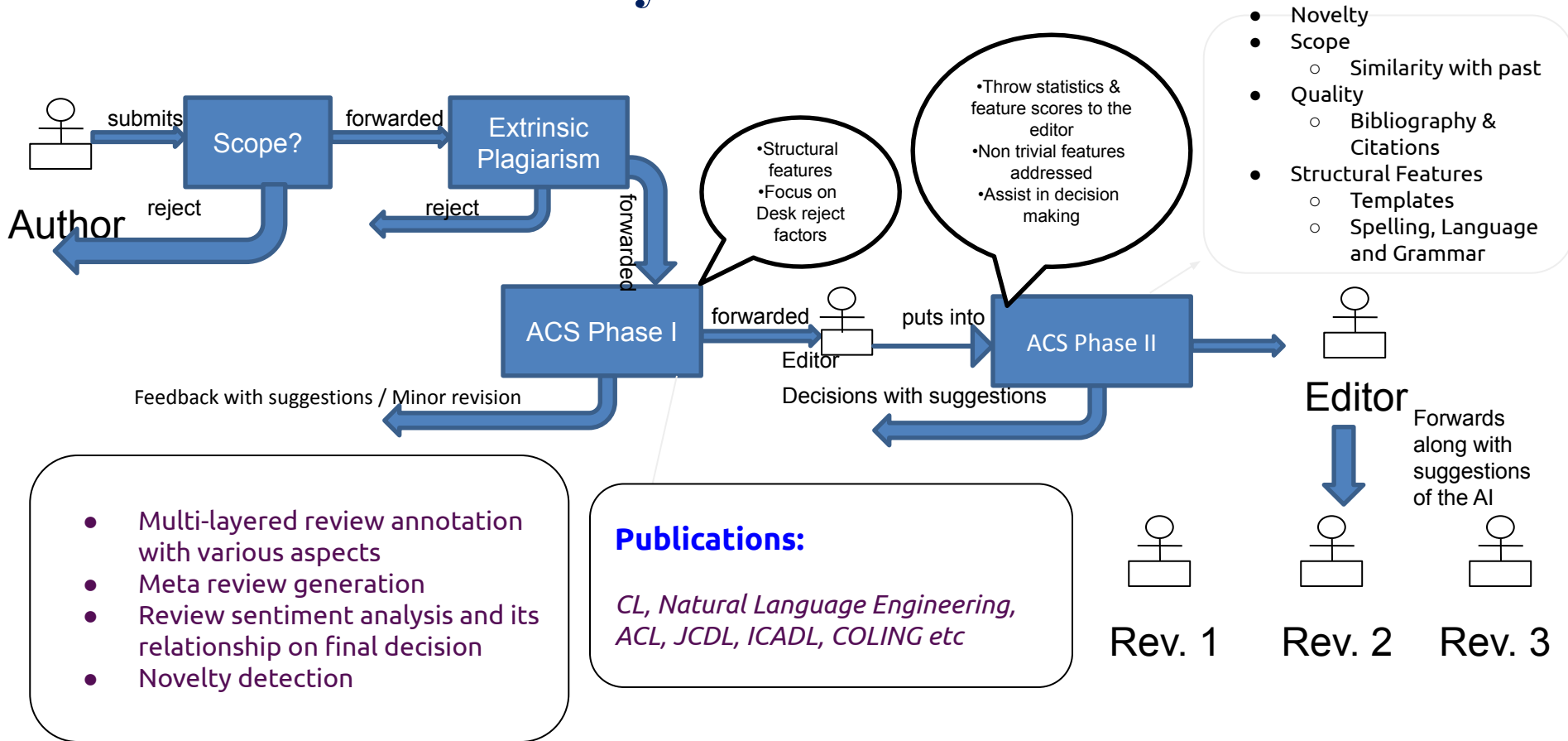
- **Peer Review**

- the only accepted method for research validation: more than 300 years old
- Old, Biased, Time-Consuming, Arbitrary
- **NIPS 2014 experiment:** Review Inconsistencies among reviewers

- *How could AI have an impact to the peer review system? How could we accelerate the peer review process?*

- Studies show that 25-50% of desk-rejections are due to papers being out-of-scope
- Extensive study done on 7500+ rejected papers reveal: **Novelty**, **Scope** and **Quality** to be the most important factors in the review process
- Around 29% of the web content is duplicate (according to a certain Google SEO Study)
- **Silver Lining:** Open Peer Review, Coalition S, FORCE11, Open Access Movement

Article Classification System & Research Facets



Machine Translation

- **Focused Areas & Techniques**

- Low-resource MT
- Unsupervised multi-lingual NMT
- SMT phrase pair injection into NMT
- Interactive NMT (*obtaining feedbacks from users after deployment*)
- Data Augmentation & Sentiment preservation in MT

- **Domains**

- Judiciary, Education, E-commerce

- **Relevance**

- Huge amount of legal documents produced daily in different languages
- Judicial documents usually very long
- Manual translation is very expensive
- ***Justice delayed is justice denied*** : Long time spent in translation
- To make educational contents available in vernacular languages (***New Education Policy***)
- Increased reliance on online shopping

Publications

*NLE, MT, ACL, EAMT,
MT Summit, ACM
TALLIP*

Machine Translation

1. **MT in judicial domain** (*sponsored by MeITY*)

- Huge amount of legal documents produced daily in different languages
- Manual translation is very expensive and time-consuming
 - **Causes significant delay in the whole judiciary process**

Justice delayed is justice denied

- English to Hindi and Hindi to English MT
 - URL for web interface is : <https://hemat.in.ngrok.io/>

2. **MT in e-commerce domain** (*sponsored by Flipkart*)

- Translating product reviews from English to Indian languages
- Demo: <https://reviewmt.in.ngrok.io/>

Conversational System: Chatbot

SEVAK-An Intelligent Indian Language Chatbot



Features

- a. Multilingual Chatbot
- b. **Languages:** English, Hindi, Bengali, Telugu and Code-mixed Languages
- c. **Domains:** Railways, Healthcare, Judiciary
- d. Integrated to **WhatsApp, Facebook** and availability via **Web Interfacing**

*An Imprint 2A, Govt of India Funded Project
(Wipro as an industry partner)*

AI for Social Good

Hitesh Golchha, Mauajama Firdaus, Asif Ekbal, Pushpak Bhattacharyya (2019). *Courteously Yours: Inducing courteous behavior in Customer Care responses using Reinforced Pointer Generator Network*. In NAACL-HLT 2019, PP. 851-860

What we solve?

To **transform** a **generic chatbot response** into a response which uses **courteous phrases** and make the users **more engaged in conversation**

Domain: Customer Care on Twitter

For any goal-oriented or open-domain systems, ***courteous response*** plays an important role in keeping the ***users engaged with the system***

Purpose is to increase user satisfaction and to build customer relations

Courtesy-The behaviour: Derived from the Politeness Theory [Brown and Levinson (1987)]

- The showing of ***politeness*** in one's attitude and behaviour towards others
- A courtesy is a polite remark or respectful act- ***very common communicative behavior***

For example,

Complain about a bad meal, and you are asked to leave ***BUT***,

the common courtesy is usually an ***apology from the manager*** and, if you're ***lucky***, a free ***dinner***

Let's see:

Some use cases of courteous behaviours in Customer Care Systems

Use-cases of Courteousness

Generic	Courteous	Behaviour
<i>How can we help?</i>	<i>Help has arrived! We are sorry to see that you are having trouble, how can we help?</i>	<i>Apology</i>
<i>Can you send us a screenshot of what you're seeing?</i>	<i>Hey Craig, help's here! Can you send us a screenshot of what you're seeing?</i>	<i>Greet</i>
<i>Let's discuss it in GM.</i>	<i>We want to help. Let's discuss it in GM.</i>	<i>Assurance</i>
<i>What is happening with your internet?</i>	<i>Oh no that's not good. I can help! What is happening with your internet?</i>	<i>Empathy</i>
<i>Enjoy your show while flying!</i>	<i>Thanks for your kind words and enjoy your show while flying!</i>	<i>Appreciation</i>

Politeness Strategies

- **Less Polite Strategies**

- seeking agreement
- joking
- expressing optimism

- **More Polite Strategies**

- being apologetic
- minimizing the imposition
- appreciating
- thanking

Resource Creation: Data Source and Attributes

- **Twitter dataset from Kaggle**

- Interactions between customers and professional customer care agents of companies
- Tweets have company names, anonymized user ids, time stamps, and response tweet ids

- **Pre-processing**

- Segment the tweet into sentences
- Remove purely *courteous (and non-informative)* sentences
- Retain purely informative sentences
- Transform the informative sentences with courteous expressions (*to remove only the courteous part from the sentence*)

Challenges

- Identifying *different variations and styles of courteous behaviours across different companies, service providers, demographics and cultures*
- Identifying courteous behaviours in the customer care domain is not straightforward
- Hard to model the **emotion across the conversation** for effective courteous response generation
 - needs to capture the **correct emotion**; and
 - accordingly handle the customers by replying courteously

For e.g.,

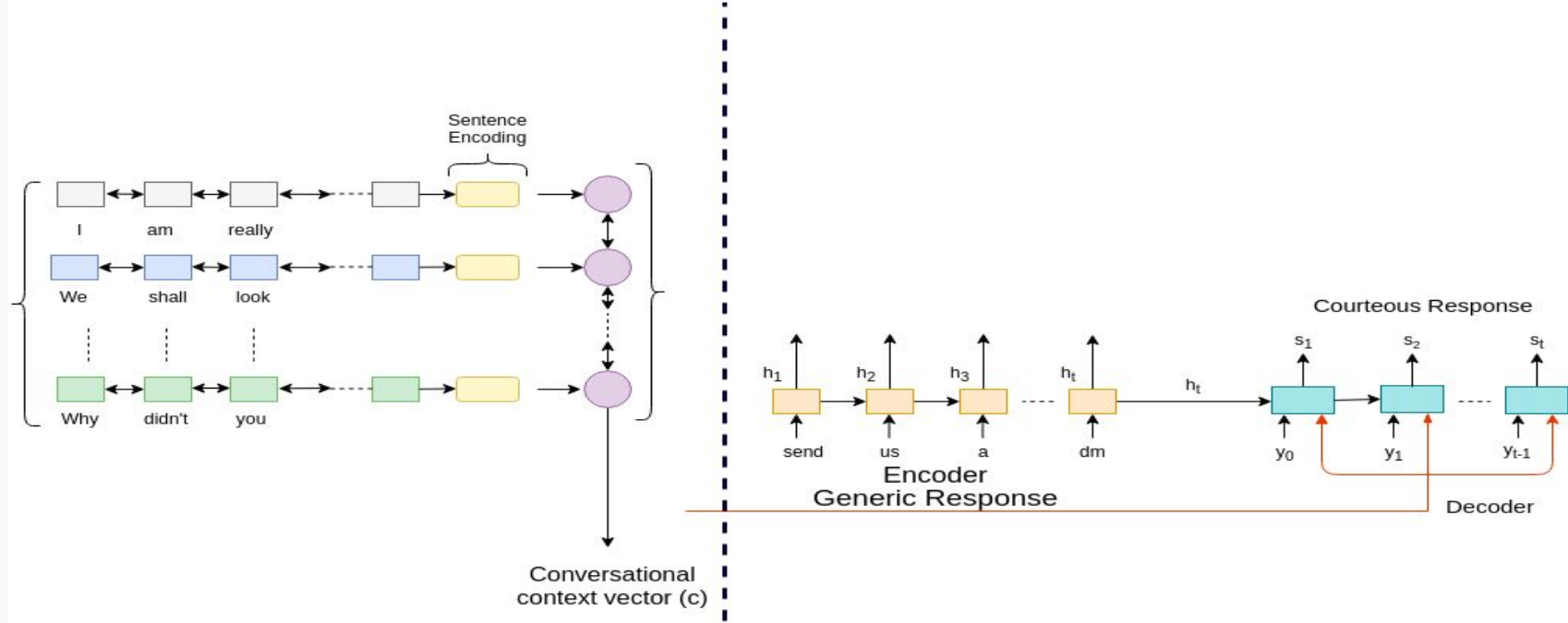
*if the customer is **angry**, the system needs to **pacify** and **apologize***

*If they are **happy**, then **appreciate***

Dataset: Statistics

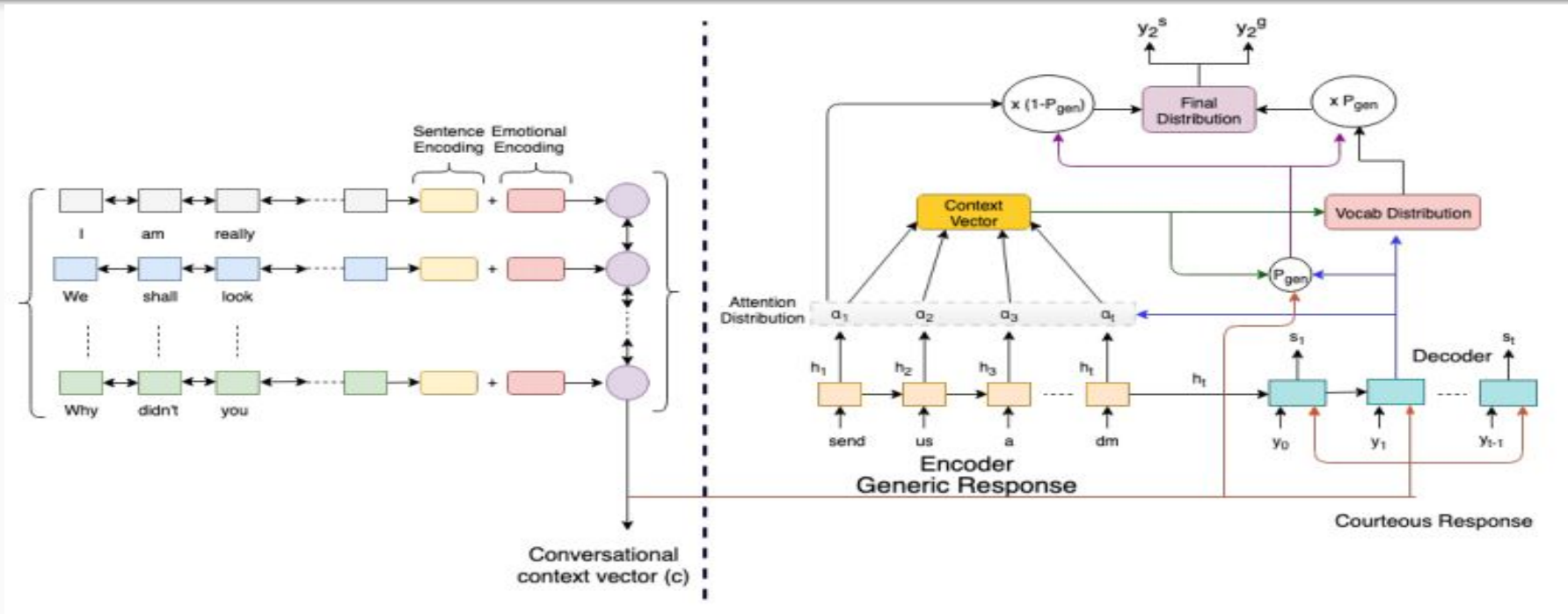
	Train	Validation	Test
# Conversations	140203	20032	40065
#Utterances	179034	25642	51238

Baseline Model



Input to the Model: Generic Response and Conversational History, **Output:** Courteous Response

Proposed Methodology



Inputs to the model: Conversation history (left), Generic response (centre), **Output:** Courteous response (right). The Conversation history is encoded by hierarchical Bi-LSTM to a Conversational Context vector c . The encoder encodes the Generic Response into hidden states h_i . Response tokens are decoded one at a time. Attention α_i , and vocabulary distributions (p_{vocab}) are computed, and combined using p_{gen} to produce output distribution. Sampling it yields y_i^s and taking its argmax yields y_i^g .

Evaluation Metrics: Automatic

- **Content Preservation**

- Measures how much of the informative content from the original generic response is reflected in the generated courteous response

- **Emotional Accuracy**

- Measures the consonance between the generated courteous expressions (source of emotion) and the gold
- Determine the similarity between the emoji distributions of the two responses

Evaluation Metrics: *Human*

- **Fluency:** Measures whether courteous response is grammatically correct and is free of any errors
- **Content Adequacy:** Measures whether the generated response contains the information present in the generic form of the response and there is no loss of information while adding the courteous part to the responses
- **Courtesy Appropriateness:** Courtesy part added to the generic responses is in accordance to the conversation history
- Scoring scheme for fluency and content adequacy:
 - 0-incorrect or incomplete; 1: moderately correct; 2: correct
- Scoring scheme for courtesy appropriateness
 - 1: in-appropriate; 0-non-courteous; 1: appropriate

Evaluation Results- *Automatic*

Model	BLEU	ROUGE			PPL	CP	EA
		1	2	L			
Seq2Seq	56.80	63.8	59.06	64.52	58.21	68.34	82.43
Seq2Seq + P	66.11	69.92	64.85	66.40	42.91	77.67	81.98
Seq2Seq + P + EE	68.16	72.18	67.92	71.17	43.52	76.05	85.75
Proposed Model	69.22	73.56	69.92	72.37	43.77	77.56	86.87

P: Pointer Generator Model; EE: Emotional embedding; PPL: Perplexity; CP: Content Preservation; EA: Emotion Accuracy

• Observations:

- Model-2 is aided by copying mechanism, and hence performance is improved as it can copy portions from generic response and forward to courtesy
- Model-3 improves the performance by 3.77 (EA) over Model-2 as it can better understand the emotional states and generate more courteous responses
- Perplexities in Model-3 and Model-4 are more compared to Model-2: may be due to emotional embeddings that confuse generated response from the ground truth

Evaluation Results

Model	F			CA			CoA		
	0	1	2	0	1	2	-1	0	1
<i>Seq2Seq</i>	15.70	42.50	41.80	16.21	41.69	42.10	23.71	51.08	25.21
<i>Seq2Seq + P</i>	14.23	42.77	43.00	15.62	39.65	44.73	22.05	39.43	38.52
<i>Seq2Seq + P + EE</i>	11.15	44.10	44.75	13.66	41.12	45.22	15.23	41.22	43.55
<i>Proposed Model</i>	10.05	44.90	44.60	13.85	38.48	47.67	14.11	41.11	44.78

F: Fluency, CA: Content Adequacy and CoA: Courtesy Appropriateness

Mental Health: *Need vs. Access*

Access to mental health care
is **poor** across the globe

- Low-income, middle-income countries
 - 1 psychiatrist per 100k individuals
- United States
 - 60% of the counties do not have a single psychiatrist

Key: We may **never** have enough mental health professionals to meet the **need**

Online peer support platforms can help!

- E.g. TalkLife, 7cups, Mental health subreddits
- Millions of users **seek** and **provide support** through **conversations**

Source: WHO's Mental Health ATLAS 2020

Mental Health and Legal Counseling for Victims (*Women & Children*): **Need**

There is a rise in the number of crimes against women and children

A counseling dialogue system designated for mental health and legal counseling of crime victims can be of great significance

- One-third of the women worldwide have experienced physical and/or sexual violence
- One billion children aged 2-17 have faced some form of violence
- Violence significantly affects their mental health
 - 20% of the global population suffers from mental health problems
- Unaware of legal and human rights
 - Lag in disclosing and reporting the assault/abuse



Counseling Conversation



Victim

I am struggling with my life after that incident and am scared to share it with anyone.



Counselor

Don't worry, could you tell which incident made you feel so?

Generic Response

During counseling, the *Counselor* should employ different counseling strategies as per ongoing conversations and *Victim's* state.



Victim

I am struggling with my life after that incident and am scared to share it with anyone.



Counselor

Don't worry, you are talking on a **safe and secure platform**. Could you tell which incident made you feel so?

Incorporation of appropriate counseling strategy (*confidentiality assurance*)

Politeness and Empathy in Counseling



Politeness

- Shows the **caring behavior**
- Improves the **sense of empathy**
- Facilitates **revelation of personal information**

([Lucas et al. 2014](#), [Kim et al. 2018](#))

Empathy

- **Core component** of counseling
- Promotes effective **therapeutic bonding** and **rapport building**
- Strong associations with **positive counseling outcomes** ([Norcross 2002](#), [Elliott et al. 2018](#))

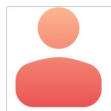
Use of *polite* and *empathetic* language

- creates familiar, warm and comfortable atmosphere to share their feelings and problems
- helps to console users and gain their trust



Please don't be sad, I understand how you feel.

Politeness and Empathy in Counseling – *An Example*



Victim

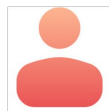
I am struggling with my life after that incident and am scared to share it with anyone.



Counselor

Kindly don't get scared, **we are here to help you** in a safe and secure environment. **Please feel free** to share with us which incident made you feel so.

Communication of *Politeness* during counseling



Victim

I am struggling with my life after that incident and am scared to share it with anyone.



Counselor

We are really sorry to hear about this, <user>. **We are here to help you** in a safe and secure environment. **We would love to know more about your problems** to assist you better, **would you be comfortable** sharing which incident made you feel so?

Communication of *Politeness* and *Empathy* (compassionate emotion) during counseling



For counseling dialogue systems to provide more effective support, the conversational agent should behave politely and empathetically.



Politeness, Empathy and Persuasiveness- *Three in One*



Victim

I am getting anxiety attacks after that incident. I often panic and feel restless. Help me.

Counselor



~~Don't worry and tell us which incident you are talking about?~~

Generic Response



We understand that anxiety attacks are scary and painful. Don't worry, **we will surely help**. A small step can relieve you from all pains, trust us and **please let us know** which incident you are talking about?

Polite, Empathetic and Persuasive Response in Counseling

Use of *empathetic*, *polite* and *persuasive* language by the counselor

- helps to console users and gain their trust
- creates familiar, warm and comfortable atmosphere to share their feelings and problems
- assists in convincing the users and change their attitudes and beliefs in their own best interest

Data Preparation: *Sources & Guidelines*

- **Sources**

- National Cybercrime Reporting Portal
- National Commission for Women
- Ministry of Women and Child Development
- Criminal Law Amendment Act 2013
- Information Technology (Amendment) Act 2008
- Real-life stories of crimes against women and children are gathered from multiple websites

- **Guidelines**

- Identifying the problem
- Building rapport with victims
- Facilitating shift in victim's circumstances and emotional state
- Providing support and safety tips (legal information, services etc)

Data Preparation: *Broader Guidelines*

- **Build relationship:** to establish a harmonious relationship with the victims by demonstrating a caring and amicable attitude towards them
- **Identify the problem:** to recognize the difficulties faced by the victim and understand the root causes of these difficulties
- **Facilitate change:** utilize suitable tactics and interventions to foster a constructive shift in victim's circumstances and emotional state
 - Imbibe politeness in the bot's responses to uncover the victim's innate qualities and maintain a seamless flow of conversation during counselling
 - Demonstrate care, warmth, and compassion to nurture a supportive environment and cultivate an empathetic bond with the victim
 - Demonstrate comprehension of the victim's experiences and emotional state inferred from their responses, in order to foster a connection between the bot and the victim, and eventually facilitate the disclosure of concealed emotions and experiences
 - Maintain a tone of affirmation and positivity in the bot's responses to uplift the victim's self-esteem

Data Preparation: *Broader Guidelines*

- **Evaluation and Termination**

- Agent should offer essential mental health and legal information/services
- contact details of experts, medical care facilities, legal guidance, information about support groups, assistance with filing complaints, etc.) to the victims, when necessary
- Agent should provide a set of safety tips to help prevent similar problems in the future

Polite and empathetic mental health and legal counseling conversational agent (AAAI, 2023)

Polite and Empathetic Counseling



I am struggling with my life after that incident and am scared to share it with anyone.

~~Don't worry, can you tell which incident made you feel so?~~

Generic Response

~~Don't worry, you are talking on a safe and secure platform. Can you tell which incident made you feel so?~~

Counseling-oriented Response

~~Kindly don't get scared, we are here to help you in a safe and secure environment. Please feel free to share with us which incident made you feel so.~~

Counseling and politeness-oriented Response

We are really sorry to hear about this, <user>. We are here to help you in a safe and secure environment. We would love to know more about your problems to assist you better, would you be comfortable sharing which incident made you feel so?

Counseling, politeness and empathy-oriented Response



What we do here?

- A **M**ental **H**earth and **L**egal **C**ounseling **D**ialogue dataset - **MHLCD**
- A **P**olite and **E**mpathetic **M**ental **H**earth and **L**egal **C**ounseling **D**ialogue **S**ystem (**Po-Em-MHLCD**)
 - Utilized reinforcement learning approach (RL)
 - Design an efficient reward consisting of
 - Task-specific rewards: **Counseling** , **Politeness**, and **Empathy**
 - Generic rewards: **Contextual-Coherence** and **Non-repetitiveness**
 - Rewards helps to generate non-repetitive and contextually-coherent responses with correct counseling-strategy, politeness, empathy imbibed in it

MHLCD Dataset

- **# conversations- 1006**
- ***Between two humans***
 - one acts as a *Counselor* (**agent**) and other as a *Victim* (**user**)
- **# Counseling Strategies- 11**
 - counselor's utterances are grounded in one of the 11 counseling strategies
- **3 politeness and 2 empathy labels**
 - counselor's each utterance is annotated with one of these

Metrics	Train	Dev	Test
<i>No. of Dialogues</i>	755	100	151
<i>No. of Utterances</i>	20886	2795	4163
<i>Average no. of utterances per dialogue</i>	27.66	27.95	27.57

Dataset Statistics

Counseling strategies, empathy and politeness labels

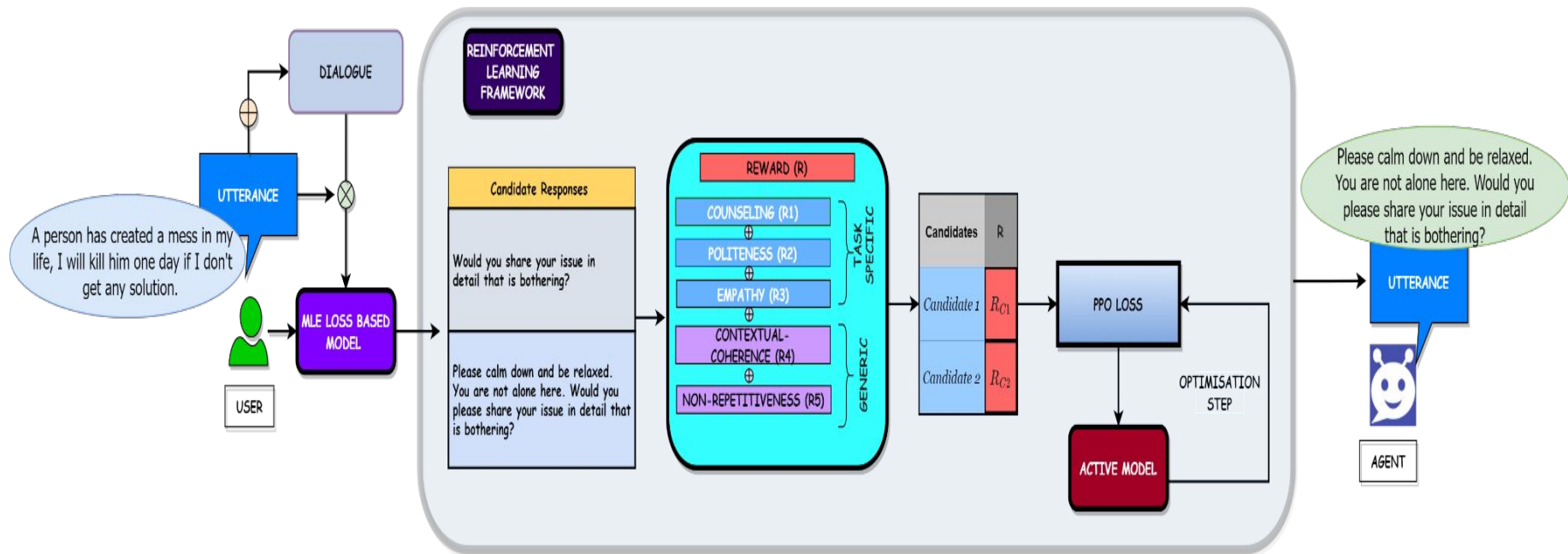
- **Counseling strategies**

- problem assessment
- confidentiality assurance
- motivational directive
- emotional support
- counseling support
- reassurance
- legal awareness
- reporting assistance
- safety guidance
- credibility assurance, and
- no-strategy

- **Politeness categories:** polite, neutral, impolite

- **Empathy categories:** empathetic and non-empathetic

Proposed System: *Po-Em-MHLCDS*



- Fine-tuned a Maximum Likelihood Estimation loss based model
 - Designed an efficient reward function consisting of sub rewards: *Task-specific rewards* viz. **Counseling**, **Politeness**, and **Empathy**; *Generic rewards* viz. **Contextual-Coherence** and **Non-repetitiveness**

Evaluation Metrics



Classifiers

- **Weighted Accuracy (W-ACC):** measures weighted accuracy of a classifier, considering all classes
- **Macro F1:** to account for imbalanced class distribution

Po-Em-MHLCDS

Automatic Evaluation

- **CoStr:** no. of utterances generated with counseling strategy
- **Pol:** no. of polite utterances generated
- **Emp:** no. of empathetic utterances generated
- **PPL:** measure the perplexity of dialogue agent's utterances
- **R-LEN:** no. of tokens in the generated utterance

Human Evaluation

- **Con:** checks for counseling strategy correctness
- **Pol, Emp:** evaluate politeness and empathy of generated responses
- **Const, Fluen, N-Rep:** to evaluate whether the generated utterances are consistent, linguistically fluent and non-repetitive in nature

Evaluation Results- *Po-Em-MHLCDS*

Models	CoStr (%)	Pol (%)	Emp (%)	PPL	R-LEN
ARDM (Wu et al. 2021)	75.24	89.10	41.30	3.21	16.02
Po-Em-MHLCDS-R	77.13	90.10	42.60	2.87	16.91
Po-Em-MHLCDS	80.30	92.54	46.40	1.91	18.71

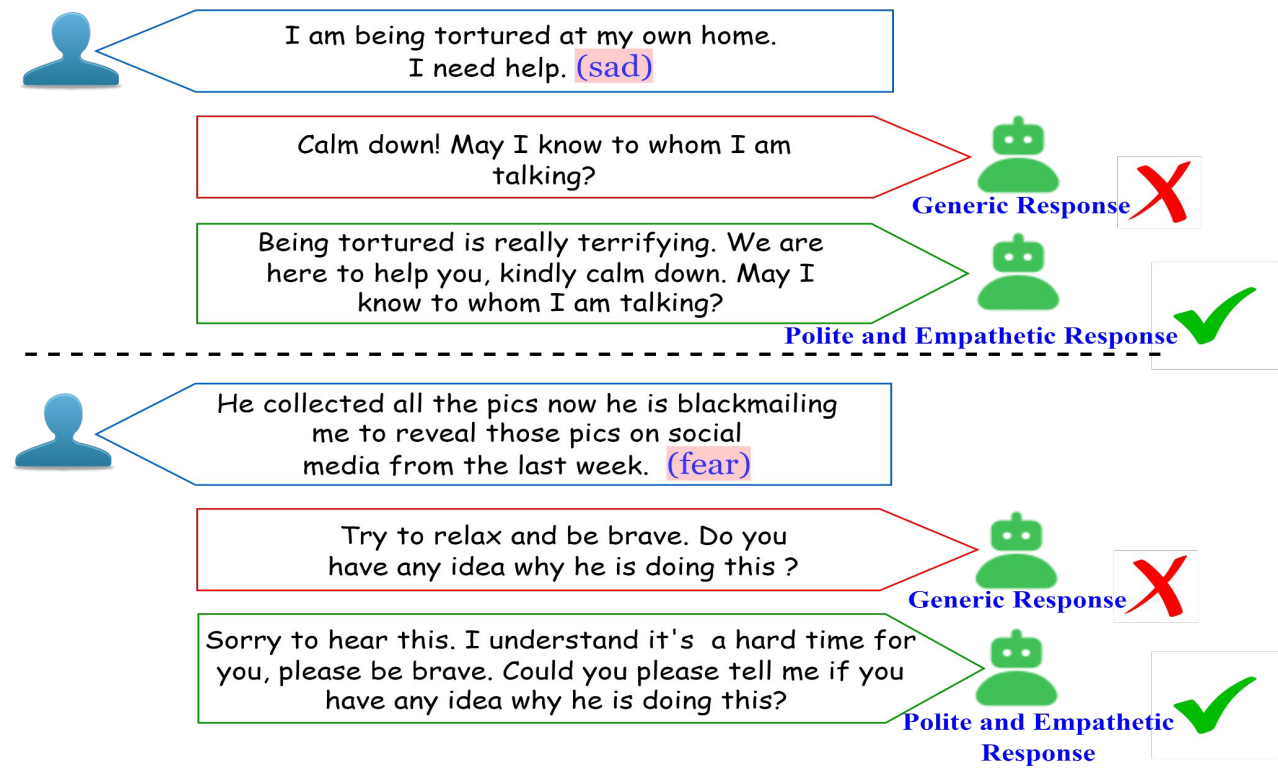
- High **CoStr**, **Pol**, and **Emp** scores show that counseling strategy, politeness and empathy rewards assist the RL-agent to generate polite and empathetic responses grounded in correct counseling strategy
- Low **PPL** and longer **R-LEN** suggest generation of
 - contextually adequate and fluent responses
 - interactive and engaging responses

Human Evaluation Results – *Po-Em-MHLCDS*

Models	Con	Pol	Emp	Const	Fluen	N-Rep
ARDM (Wu et al. 2021)	3.04	3.83	2.13	3.74	4.12	3.87
Po-Em-MHLCDS-R	3.39	3.96	2.28	3.91	4.31	4.11
Po-Em-MHLCDS	3.94	4.41	2.85	4.16	4.57	4.72

- High **Con**, **Pol**, **Emp**, **Const**, **Fluen**, and **N-Rep** scores show that contextual-coherence and fluency rewards enable the model to generate consistent, fluent and non-repetitive utterances
- Politeness and empathy rewards with counseling reward help **Po-Em-MHLCDS** in
 - generating engaging and interactive responses
 - building a rapport with the victim

Emotion-adaptive polite and empathetic conversational agent (ACL, 2023)



What has been done here?

- Counseling conversational dataset annotated with emotion, politeness and empathy labels - (**EPE-enEIH**)
- A novel emotion-adaptive **P**olite and emp**A**thetic counse**L**ing conversational agent (**PAL**)
 - Utilized reinforcement learning approach (RL)
 - Design an efficient reward consisting of
 - Preference rewards: **Utterance-emotion-politeness Consistency**, **Utterance-emotion-empathy Consistency**, **Politeness-adaptive**, **Empathy-adaptive**, **Politeness Correctness**, **Empathy Correctness**
 - Generic rewards: **Dialogue Flow Consistency** and **Diversity**
 - Rewards helps to to generate engaging, fluent and interactive client's emotion-adaptive polite and empathetic responses

Persuasiveness and Politeness for Empathetic conversational agent (IJCAI, 2023)



Victim

I am getting anxiety attacks after that incident. I often panic and feel restless. Help me.

Counselor

Generic Response

Don't worry and tell us which incident you are talking about?



Politeness Strategy-adaptive Response

Don't worry, ~~we will surely help. Could you please~~ tell us which incident you are talking about?



Politeness and Empathy Strategy-adaptive Response

~~We understand that anxiety attacks are scary and painful. Don't worry, we will surely help. Can you please~~ let us know which incident you are talking about?



Politeness and Empathy Strategy-adaptive Persuasive Response

We understand that anxiety attacks are scary and painful. Don't worry, ~~we will surely help.~~ A small step can relieve you from all pains, trust us and please let us know which incident you are talking about?



HEAL Dataset

- **# conversations-** 216
- Between two humans
 - one acts as a Counselling bot (agent) and other as a Victim (user)
- **# Counseling Acts-** 8
 - counselor's each utterance is grounded in one of the 8 counseling acts
- **# Empathy Strategies-**7
 - counselor's each utterance is grounded in one of the 7 empathy strategies
- **# Politeness Strategies-**3
 - counselor's utterances are grounded in one of the 3 politeness strategies (**Brown et al. 1987**)

Metrics	Train	Dev	Test
<i>No. of Dialogues</i>	162	22	32
<i>No. of Utterances</i>	4133	407	664
<i>Average no. of utterances per dialogue</i>	25.51	18.50	20.75

Dataset Statistics

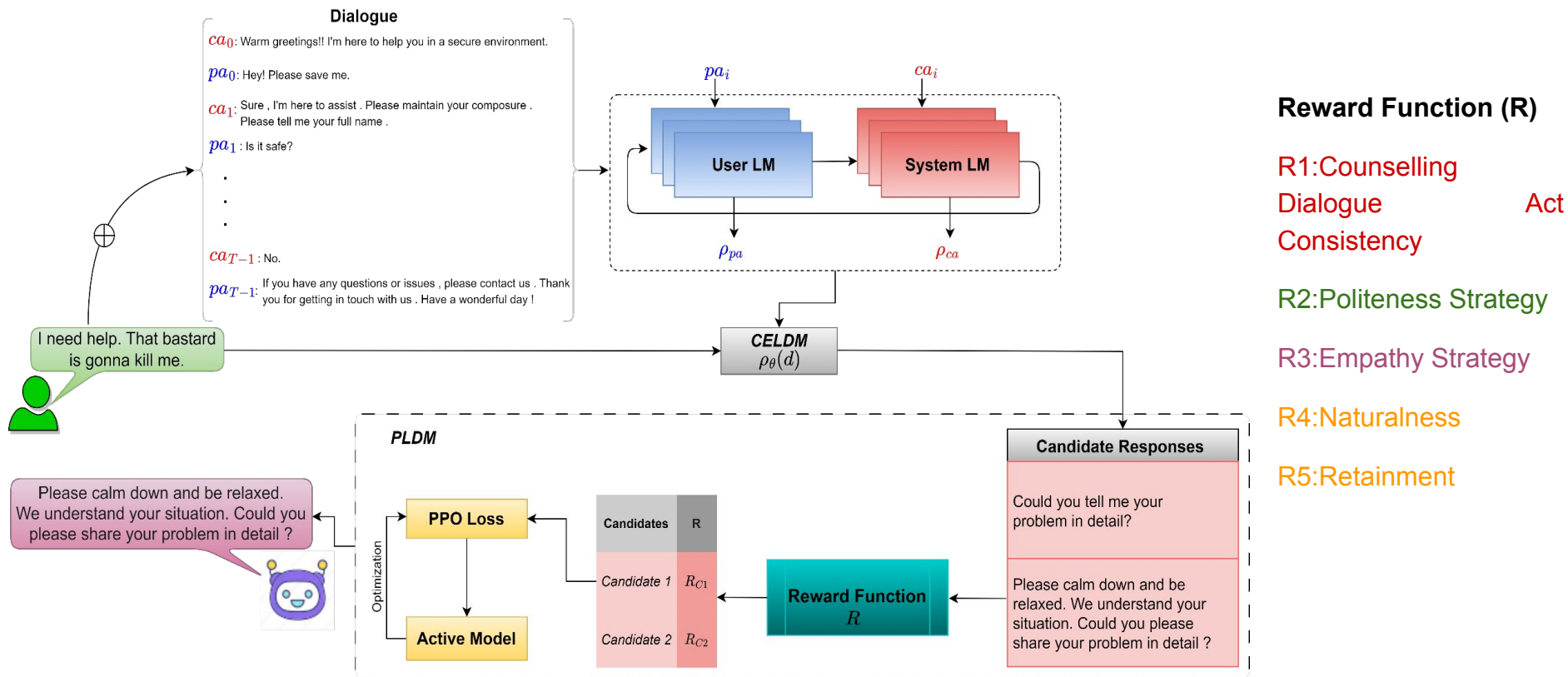
Counselling Act- *meeting the adequate requirement of victims*

Act	Purpose
Counselling support	Provide various support services like medical help, mental health-related aid, NGOs information etc. during counselling based on the victim's need
Legal assistance	Ensures legal assistance to the victims
Persuasion	Assists the victims in developing a readiness to seek professional assistance by compelling them to adhere to specific recommendations, modifying their attitudes and beliefs in their own best interests, and fostering a sense of readiness to do so
Seek information	Request for a few basic information in order to comprehend the problem and provide relevant assistance
Deliver information	Provide information pertaining to the problem being discussed
Re-check assistance	Inquires for further help or clarification about the problem under discussion
Greet	Typically, each conversation begins with a greeting from one speaker and an appropriate response from the other
Closing remark	Marks the end of the conversation

Empathetic Strategies– *to establish a personal, friendly, and empathetic connection with the victim*

Strategy	Purpose
Reflective listening	Demonstrates a genuine curiosity to learn and delve deeper into the details shared by the victims, creating a sense that the bot is genuinely interested in listening to them
Confidential comforting	Displays genuine interest and concern for the privacy of the victims, providing assurance that any information shared will be treated with the utmost confidentiality
Evoke motivation	Encourages the victims to embrace a forward-looking perspective and participate in activities that promote feelings of positivity and optimism
Express emotional support	Provides emotional solace or words of encouragement to entirely comprehend the problems faced by the victims and the intensity of their emotions
Offer counselling	Provides essential mental health and legal counselling advice, along with contact information of experts, whom the victim can reach out to for further guidance and assistance.
Escalate assurance	Reassures the victims that they are never to blame for any form of assault and firmly conveys the message that they are not alone, emphasizing that they can always seek help and support.
No strategy	Assigned to the utterances which do not utilize any empathy strategy

PARTNER



Evaluation Results- *Automatic*

Models	CoAct (%)	EmpStr (%)	PolStr (%)	PPL	R-LEN
ARDM (Wu et al. 2021)	52.80	57.20	66.10	3.74	14.8
PARTNER-R	51.90	57.30	66.30	3.68	14.4
PARTNER-GR	55.30	59.10	69.30	3.17	15.7
PARTNER-TR	53.80	58.20	67.80	3.31	15.4
Po-Em-MHLCDS	56.50	61.80	69.9	2.55	16.06

- High **CoAct**, **EmpStr**, and **PolStr** scores depicts that PARTNER generate appropriate counselling act, empathy, and politeness strategy-adaptive responses
- Low **PPL** and longer **R-LEN** suggest generation of contextually adequate and fluent responses

Evaluation Results – *Human*

Models	CounC	EmpC	PolC	Nats	Corr	NRep
ARDM (Wu et al. 2021)	2.44	2.60	2.97	4.10	3.84	3.71
PARTNER-R	2.36	2.66	3.01	4.21	3.91	3.82
PARTNER	3.06	3.11	3.42	4.52	4.14	4.07

High **CounC**, **EmpC**, **PolC**, **Nats**, **Corr**, and **NRep** scores depicts that PARTNER is able to generate natural , accurate, and non-repetitive counseling utterance utilizing correct politeness and empathy strategies

Priya et al. PARTNER: A Persuasive Mental Health and Legal Counselling Dialogue System for Women and Children Crime Victims. In IJCAI 2023

Representative List of Projects

- Centre of Excellence for Natural Language Processing-**Reed Elsevier Publishing Company**
 - Some novel R&D problems in the broad areas of AI, NLP and ML (AI Assisted Peer Review System, Email Classification etc.)
- **Conversational Agents with Negotiation and Influencing ability- Accenture**
 - **Research on negotiation and persuasive chatbot**
- MeiTY, Govt. of India (with IITB, IIT Delhi, IIITH, IIITA, CDAC Pune)
 - **Hindi to English Machine Translation for Judicial Domain**
- Imprint India, MHRD (with IIT Kharagpur, DAIICT, Au-KBC and IITB)
 - **A Platform for Cross-lingual and Multilingual Event Monitoring in Indian Languages**
- **Percuro: A Holistic Solution for Text Mining (with AISC, USA; *sponsored by Wipro*)**
 - QA, Summarization, KG
- **HELIOS - Hate, Hyperpartisan, and Hyperpluralism Elicitation and Observer System, Sponsored by Wipro Ltd, I**
(**Consortium partners: UT Austin, USA; and IIIT Delhi, India**)
 - Hate Speech Detection, Fake News Detection

Representative List of Projects

- *Vidyaapati*: A System for Bidirectional Machine Translation Involving Bengali, Konkani, Maithili, Marathi and Hindi, NLTM, MeITY, Govt of India (*with IITB, JNU, CDACP, CDACK, JU, ISI Kolkata*)
 - Neural Machine Translation
 - Low-resource Languages
- Bhashini: Indian Language to Indian Language Machine Translation System, Sponsored by NLTM, Meity, Govt of India with IIITH, DAIICT, MIT, CDAC Noida etc)
 - Machine Translation from IL-to-IL
 - Evaluation Schema
- IITP Centre of Excellence in Cyber Crime Prevention against Women and Children-AI based Tools for Women and Children Safety, sponsored by Ministry of Home Affairs (MHA)
 - *Multimodal offensive content detection, Counseling Chatbot*

***Thank you for your
attention!***